FIT3080 Applied class: Supervised ML & Artificial Neural Networks

Exercise 1: ***Artificial Neural Network***

Below is a diagram of a single artificial neuron (unit),

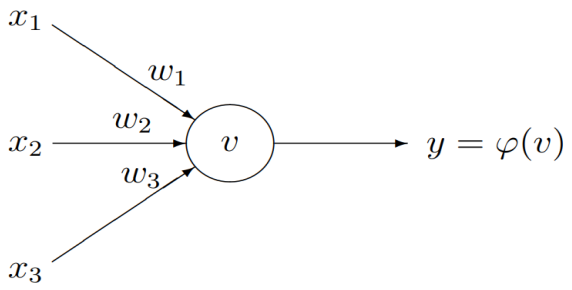


Figure 1: Single unit with three inputs.

The node has three inputs x = (x_1, x_2, x_3) with w = (w_1, w_2, w_3) respectively. This node receives only binary signals (either 0 or 1). Suppose that the weights corresponding to the three inputs have the following values:

w_1	= 2
w_2	= - 4
w_3	= 1

And the activation of the unit is given by the step function:

$$\varphi(v) = \{1 \quad if\ v \geq 0;\ 0 \quad otherwise\ \}$$

Calculate what will be the output value *y* of the unit for each of the following input patterns:

Input Pattern	P_1	P_2	P_3	P_4
x_1	1	0	1	1
x_2	0	1	0	1
x_3	0	1	1	1

Exercise 2*: Linear Regression*

A linear regression model is used to predict the price of a used car based on its age.

a) Write the general form of a simple linear regression model and explain its parameters.

b) Suppose the model $\hat{y} = 30 - 3x$, where x is the car's age in years and y is its price in thousands of dollars. For two cars aged 2 and 4 years, the actual prices are $25,000 and $20,000. Calculate the predicted prices and squared errors.

c) Write the Sum of Squared Errors (SSE) for these two cars and its general formula for n observations.

Exercise 3: *Regression via Backpropagation*

A small dataset relates hours studied (X) to exam scores (Y):

Hours Studied (X)	Exam Score (Y)
1	40
2	50
3	65

We want to fit a simple linear regression model:

$\hat{y} = a + bx$.

Perform two-iteration gradient descent updates with learning rate $\alpha=0.1$ (backpropagation). Assume the initial parameters are $a = 0$ and $b = 0$.

Exercise 4: ***Logistic Regression***

A teacher wants to predict whether students pass (Y=1) or fail (Y=0) an exam based on hours studied (X). The dataset is:

Hours Studied (X)	Pass/Fail (Y)
1	0
2	0
3	1
4	1

We use the logistic regression model:

$$P(Y=1|X) = \sigma(a + bX), \quad \sigma(z) = 1 / (1+e^{-z})$$

a) Write down the squared-error loss function used for logistic regression in the lecture. Use $L = (1/2) \Sigma (Y_i - p_i)^2$, where $p_i = \sigma(a + bX_i)$.

b) With initial parameters a=0, b=1, compute the predicted probabilities for each data point.

c) Perform two full-batch gradient descent iterations with learning rate $\alpha = 0.1$, starting from a = 0 and b = 1. Use the weight-update rule from the lecture (slide 25), summing contributions from all four data points at each iteration. Compute the predicted probabilities after each iteration.

Exercise 5: ***Convolutional Neural Network (Optional)***

You were contacted by a food processing company that wants you to develop a classifier that detects whether a rat is present in an image. You collect a large dataset of images by crawling the web, and have annotators determine which images contain rats. This set of images can then be used as the training set for your classifier.

a) Discuss whether CNN is suitable for this classification task.

b) After you have delivered your solution to the company, they get back to you and complain that when they evaluate on a new test set, they get precision and recall values that are much lower than what you reported to them. Explain what might have gone wrong.

c) Given that the image *R* represents a grayscale image as the input, *K* is a 2 x 2 kernel (filter). Perform the convolution below (stride =1) to calculate the feature map *C*.

	1	0	5	0	2	0
	0	1	0	0	0	9
R =	0	7	0	3	0	0
	1	0	0	0	0	1
	2	0	2	0	6	0

K =	2	3
	1	1

d) Based on the obtained feature map *C,* perform the max pooling operation (2 x 2 filter, stride =1).